

Session 10

Part A – Setup and Dataset

Q1. Import the required libraries.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_squared_error, r2_score
```

Part B – Create Dataset

Q2. Create a dataset showing student performance influenced by multiple features.

Step 1 – Create the input features (already provided):

```
data = {
    "Hours_Studied": [2, 3, 4, 5, 6, 7, 8, 9, 10, 11,
                     3, 5, 7, 8, 9, 10, 11, 6, 4, 5],
    "Attendance": [60, 65, 70, 75, 80, 85, 90, 92, 95, 98,
                  70, 75, 80, 85, 88, 92, 96, 83, 78, 74],
    "Sleep_Hours": [4, 5, 6, 6, 7, 8, 8, 5, 6, 7,
                   5, 6, 7, 7, 8, 8, 9, 6, 5, 7],
    "Assignments_Completed": [4, 5, 6, 6, 7, 8, 9, 10, 10, 10,
                              5, 6, 7, 8, 9, 10, 10, 8, 6, 7]
}
```

Step 2 – Create the random number generator and add a target column Final_Score:

```
data["Final_Score"] = (
    3 * np.array(data["Hours_Studied"]) +
    0.4 * np.array(data["Attendance"]) +
    2 * np.array(data["Assignments_Completed"]) +
    rng.standard_normal(20) * 3 # noise
)
```

(This is how final score should be)

Step 3 – Final data Frame

	Hours_Studied	Attendance	Sleep_Hours	Assignments_Completed	Final_Score
0	2	60	4	4	38.914151
1	3	65	5	5	41.880048
2	4	70	6	6	54.251354
2	4	70	6	6	54.251354
3	5	75	6	6	59.821694
4	6	80	7	7	58.146894

Part C – Train-Test Split

Q3. Split the dataset into training (70%) and testing (30%) sets.

Part D – Train Models with Different Complexity

Q4. Train a Linear Regression model.

Q5. Train a low-depth Decision Tree (simple model).

Q6. Train a high-depth Decision Tree (complex model).

Part E – Compare Model Performance

Q7. Compare all models using MSE and R^2 .

Part F – Visualize Actual vs Predicted

Q8. Plot model predictions vs actual scores.

Part F – Reflection Questions

Q9. Which model has high bias (underfitting)?

Q10. Which model has high variance (overfitting)?

Q11. Which model balances both bias and variance best?

Q12. How can you control model complexity in trees?

Part G

Q13. Train more trees with depths from 1–10 and plot R^2 scores.

Q14. Identify the optimal tree depth from the graph.